

A strengthened profile-integral maximum-degree bound for diameter-2 edge-critical graphs

Reviewer edition 1 - current strongest profile-integral candidate

Paul Lenz research project; mathematical development and drafting with ChatGPT/Geeps

9 September 2026

Abstract

We present a strengthened candidate profile-integral argument. A sharper elementary square-root minorant improves the universal surplus estimate, and an exact finite threshold certificate completes the degree assembly at the rational threshold $7/12$. Two separately written standard-library checkers agree on the scalar arithmetic and every finite exception. The argument remains candidate mathematics: independent specialist review, novelty assessment and external computational reproduction are open.

Reviewer status. complete candidate hand argument; internal exact audits green; independent review and novelty assessment OPEN. This reviewer edition is an editorial rendering of the canonical source proof listed below. It does not convert same-assistant checking into external independence, does not make a novelty or priority claim, and does not claim the unrestricted Murty-Simon conjecture unless the source proof itself proves such a statement.

Canonical claim. $n \geq 6$ and $\Delta(G) \geq (7/12)n$ imply $e(G) < \text{floor}(n^2/4)$.

Canonical proof source. `project/research/general_n/2026-09-09-profile-integral-7-12-v1/PROOF.md`. The source is reproduced in full below; substantive mathematical changes must be made in the canonical proof first and then rebuilt into this edition.

A strengthened profile-integral bound and a $7/12$ maximum-degree candidate

9 September 2026. Paul Lenz: research direction. ChatGPT/Geeps: derivation, drafting and internal checking.

Status: complete candidate hand proof; internal exact audits green. Independent mathematical review, novelty assessment and external computational reproduction remain OPEN. This is not an unrestricted solution of Murty-Simon.

Statement

Let G be a finite simple diameter-two edge-critical graph on $n \geq 6$ vertices, with m edges and maximum degree b . Put

$$\begin{aligned} a &= n-1-b, \\ t &= m-b(n-b). \end{aligned}$$

The argument proves the strengthened universal surplus bound

$$t < \frac{5a^2}{128} + \frac{a}{8}. \quad (\text{P+})$$

Together with an exact finite threshold certificate, it gives the candidate implication

$$b \geq \frac{7}{12}n \implies m < \left\lfloor \frac{n^2}{4} \right\rfloor. \quad (\text{T+})$$

The proof uses the same graph-to-demand and exact threshold-capacity lemmas as the earlier 293/500 profile-integral candidate. The new mathematical ingredient is a sharper elementary scalar estimate; no new graph bridge is assumed.

1. Shared graph-to-profile lemma

Let H be the complement of G . Choose a minimum-degree vertex v of H , put

$$A = N_H(v), \quad |A| = a, \\ B = V(H) \setminus N_H[v], \quad |B| = b.$$

Use the selected quasi-edge construction for missing pairs of $H[B]$, with residual source degrees ρ_u , residual label degrees R_i , selected label degrees x_i , and common residual total r . Let F be the complement of $H[A]$, with $d_i = \deg_F(i)$, and define

$$s_i = \max(0, d_i - R_i), \quad S = \sum_i s_i.$$

The established exact ledger and minimum-degree argument give

$$e(F) = r + t, \quad \sum_i d_i = 2(r + t), \quad S \geq r + 2t. \quad (1)$$

For a selected incidence from source u to label i , the quasi-edge covering argument gives

$$s_i \leq \rho_u. \quad (2)$$

For each integer $h \geq 1$, define

$$I_h = \{i : s_i \geq h\}, \quad W_h = \sum_{i \in I_h} s_i,$$

$$Z_h = \{u : \rho_u \geq h\}, \quad z_h = |Z_h|.$$

Counting selected orientations and their supplements gives the exact threshold-capacity inequality

$$2W_h \leq z_h^2 - z_h + h(h + 1). \quad (3)$$

If $H_0 = \max_i s_i > 0$, then a label of demand H_0 needs H_0 distinct sources of residual degree at least H_0 , and the nested source sets satisfy

$$z_h \geq H_0 \geq h \quad (1 \leq h \leq H_0), \quad \sum_{h=1}^{H_0} z_h \leq r. \quad (4)$$

These are graph-derived necessary conditions, not empirical observations.

2. The profile integral

For $h \leq H_0$, let $\ell_h = |I_h|$ and $p = \ell_h / a$. Cauchy–Schwarz gives

$$\left[\frac{1}{a} \sum_{i \in I_h} \sqrt{2as_i - h^2} \right]^2 \leq 2pW_h - p^2h^2.$$

Using (3), $0 \leq p \leq 1$, and $z_h \geq h$,

$$2pW_h - p^2h^2 \leq pz_h^2 + p(1-p)h^2 \leq z_h^2.$$

Hence

$$z_h \geq \frac{1}{a} \sum_{i: s_i \geq h} \sqrt{2as_i - h^2}. \quad (5)$$

Summing over h and using (4),

$$r \geq \frac{1}{a} \sum_i D_a(s_i), \quad D_a(s) = \sum_{h=1}^s \sqrt{2as - h^2}. \quad (6)$$

Define

$$\Phi(x) = \int_0^x \sqrt{2x - y^2} dy, \quad 0 \leq x \leq 1.$$

The shifted-midpoint concavity argument from the profile-integral proof gives, for every integer $0 \leq s \leq a-1$,

$$D_a(s) \geq a^2 \Phi(s/a) - a/4. \quad (7)$$

For completeness, if $s > 0$, put $f(y) = \sqrt{2as - y^2}$. On $[0, s+1/2]$ it is decreasing and concave; midpoint concavity gives

$$\sum_{h=1}^s f(h) \geq \int_{1/2}^{s+1/2} f(y) dy.$$

The difference from the integral over $[0, s]$ is at most

$$\frac{f(0) - f(s+1/2)}{2} \leq \frac{\sqrt{2as} - s}{2} \leq a/4,$$

because $(s+a/2)^2 \geq 2as$. Rescaling the integral gives (7).

Combining (6) and (7),

$$r \geq a \sum_i \Phi(s_i/a) - a/4. \quad (8)$$

Everything from (1) through (8) is shared with the previous profile-integral route.

3. Sharper scalar estimate

We now prove

$$x - \Phi(x) < \frac{5}{64} \quad (0 \leq x \leq 1). \quad (9)$$

3.1 A sharper square-root minorant on the actual domain

For $0 \leq u \leq 1/2$, put

$$B(u) = 1 - \frac{u}{2} - \frac{u^2}{8} - \frac{3u^3}{32}.$$

Direct expansion gives

$$1 - u - B(u)^2 = \frac{u^3}{1024} (64 - 112u - 24u^2 - 9u^3). \quad (10)$$

Both $B(u)$ and the cubic in parentheses are decreasing on $[0, 1/2]$, while

$$B(1/2) = 181/256 > 0,$$

$$64 - 112(1/2) - 24(1/2)^2 - 9(1/2)^3 = 7/8 > 0.$$

Therefore $B(u) \geq 0$ and $B(u)^2 \leq 1 - u$, so

$$\sqrt{1-u} \geq B(u) \quad (0 \leq u \leq 1/2). \quad (11)$$

For $x > 0$, in the integral for $\Phi(x)$ take

$$u = y^2/(2x).$$

Since $0 \leq y \leq x \leq 1$, one has $0 \leq u \leq x/2 \leq 1/2$; this restricted range is what allows the improvement over the previous $1/12$ estimate. Integrating (11) gives

$$\Phi(x) \geq \sqrt{2x} \left(x - \frac{x^2}{12} - \frac{x^3}{160} - \frac{3x^4}{1792} \right). \quad (12)$$

3.2 Exact polynomial bound

Put

$$q = \sqrt{x/2}, \quad 0 \leq q \leq 1/\sqrt{2}.$$

Then (12) gives

$$x - \Phi(x) \leq P(q),$$

where

$$P(q) = 2q^2 - 4q^3 + \frac{2}{3}q^5 + \frac{1}{10}q^7 + \frac{3}{56}q^9. \quad (13)$$

Write

$$P'(q) = qQ(q),$$

$$Q(q) = 4 - 12q + \frac{10}{3}q^3 + \frac{7}{10}q^5 + \frac{27}{56}q^7.$$

On $0 \leq q \leq 1/3$, $Q(q) \geq 4 - 12q \geq 0$, so P is increasing.

On $7/20 \leq q \leq 1/\sqrt{2}$,

$$Q'(q) = -12 + 10q^2 + \frac{7}{2}q^4 + \frac{27}{8}q^6 \leq -\frac{365}{64} < 0.$$

Moreover exact arithmetic gives

$$Q(7/20) = -\frac{1631127391}{30720000000} < 0.$$

Thus P is decreasing from $7/20$ onward.

It remains only the short interval $1/3 \leq q \leq 7/20$. There

$$2q^2 - 4q^3 \leq 2/27$$

because this cubic is decreasing for $q \geq 1/3$, while the positive tail is increasing and

$$\frac{2}{3}(7/20)^5 + \frac{1}{10}(7/20)^7 + \frac{3}{56}(7/20)^9 = \frac{43868404489}{12288000000000} < \frac{1}{250}.$$

Consequently, throughout all three intervals,

$$P(q) < \frac{2}{27} + \frac{1}{250} < \frac{5}{64},$$

with exact margin

$$\frac{5}{64} - \left(\frac{2}{27} + \frac{1}{250} \right) = \frac{11}{216000} > 0.$$

This proves (9), including $x=0$ directly.

4. Strengthened universal surplus bound

Apply (9) to every s_i/a in (8):

$$r > S - \frac{5a^2}{64} - \frac{a}{4}.$$

Since (1) gives $2t \leq S - r$,

$$2t < \frac{5a^2}{64} + \frac{a}{4},$$

hence

$$t < \frac{5a^2}{128} + \frac{a}{8}.$$

This proves (P+).

5. Degree assembly for 7/12

Assume now

$$b \geq 7n/12, \quad n = a + b + 1.$$

Then

$$5b \geq 7(a + 1),$$

so

$$b - n/2 = \frac{b - a - 1}{2} \geq \frac{a + 1}{5}. \quad (14)$$

If, contrary to (T+),

$$m \geq \lfloor n^2/4 \rfloor,$$

then

$$t \geq \lfloor n^2/4 \rfloor - b(n - b) = \left\lfloor \frac{(b - a - 1)^2}{4} \right\rfloor \geq (b - n/2)^2 - 1/4.$$

Using (14),

$$t \geq \frac{(a+1)^2}{25} - \frac{1}{4}. \quad (15)$$

Subtracting the upper bound (P+) from the right side of (15) gives

$$D(a) = (a+1)^2/25 - 1/4 - 5a^2/128 - a/8.$$

Its quadratic coefficient is

$$\frac{1}{25} - \frac{5}{128} = \frac{3}{3200} > 0.$$

Exact arithmetic gives

$$D(53) = 123/3200 > 0,$$

and

$$D(54) - D(53) = 177/3200 > 0.$$

Subsequent forward differences increase by $6/3200$. Hence $D(a) > 0$ for every integer $a \geq 53$, contradicting (P+).

6. Exact finite assembly for $a \leq 52$

For fixed a , the least integer b satisfying the degree premise is

$$b_0 = \left\lceil \frac{7(a+1)}{5} \right\rceil.$$

Since $b_0 > a+1$, the required surplus

$$T_0 = \left\lfloor \frac{(b_0 - a - 1)^2}{4} \right\rfloor$$

is nondecreasing for larger eligible b .

For $2 \leq a \leq 52$, the strict bound (P+) immediately contradicts $t \geq T_0$ whenever

$$128T_0 \geq 5a^2 + 16a. \quad (16)$$

Exact integer arithmetic shows that (16) fails only for

$$a = 4, 6, 9, 11, 14, 19, 24, 29, 34, 39, 44.$$

These eleven cases are closed by the same finite threshold certificate already used in the profile-integral programme. For fixed a , maximum demand $H > 0$, and total demand S , define for $1 \leq h \leq H$

$$K_h = \max \left(1, \left\lceil \frac{S - a(h-1)}{H - h + 1} \right\rceil \right),$$

$$w_h = \max \{ H, hK_h, S - (a - K_h)(h-1) \}.$$

Any demand multiset with maximum H and total S has at least K_h entries at least h , hence $W_h \geq w_h$. Let L_h be the least integer $z \geq H$ satisfying

$$z^2 - z + h(h + 1) \geq 2w_h.$$

By (3)-(4), $z_h \geq L_h$ and therefore

$$r \geq \sum_{h=1}^H L_h,$$

so

$$S - r \leq S - \sum_{h=1}^H L_h. \quad (17)$$

The exact maxima of (17) over all integer $1 \leq H \leq a-1$, $H \leq S \leq aH$ are:

a	least b	required T0	max upper bound on S-r
4	7	1	1
6	10	2	2
9	14	4	5
11	17	6	8
14	21	9	13
19	28	16	26
24	35	25	43
29	42	36	64
34	49	49	90
39	56	64	121
44	63	81	155

Every entry in the last column is strictly below $2T_0$. Since $2t \leq S - r$, each exception contradicts $t \geq T_0$. The zero-demand case $S=0$ gives $t \leq 0$ separately.

For $a=0$, edge-criticality plus a universal vertex forces a star, so $m=n-1 < \text{floor}(n^2/4)$ for $n \geq 6$. For $a=1$, F is edgeless and $t=-r \leq 0$, whereas the eligible degree premise requires positive surplus relative to the Turán threshold. Thus all cases are covered.

This proves the candidate implication (T+).

7. Exact checker and trust boundary

`src/check_7_12.py` verifies with standard-library rational/integer arithmetic:

1. the square-root minorant identity and endpoint signs;
2. the $5/64$ scalar polynomial certificate;
3. the large- a quadratic margin and forward difference;
4. completeness of the eleven finite assembly exceptions;
5. the exact threshold certificate for all eleven exceptions;
6. a degree-pair regression through $n=5000$.

The regression is not proof by exhaustion; it is a consistency check on the hand assembly.

The principal mathematical trust boundary remains the shared graph-to-demand and threshold-capacity argument in Sections 1–2. A flaw there would affect both this strengthening and the earlier profile-integral candidate. Same-assistant checking is not external review.